

Probabilistic Robotics Group 18 - Final Project Proposal

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1. Idea

This project aims to integrate what we have learned in class - robot perception and control, SLAM - and reinforcement learning into a practical and functional robot system. Using Gazebo and a TurtleBot3, the robot will build a map of an unknown environment and estimate its own position through EKF-SLAM, while performing basic motion control based on the estimated pose. Then, we will incorporate reinforcement learning so the robot can learn how to avoid obstacles and move toward a target using sensor data such as LiDAR. Through this project, we hope to better understand how SLAM, control, and reinforcement learning work together in a real robotic system, and gain hands-on experience in applying these concepts to a probabilistic robotics project.

2. Technology Stack

Operating System: Ubuntu 22.04, ROS 2 Humble, Gazebo

SLAM: EKF-SLAM for localization and state estimation, Nav2 toolbox for map generation

Reinforcement Learning: Stable-Baselines3 (PPO) integrated with Nav2 for robot obstacle avoidance and navigation training

Robot: TurtleBot3

3. Project Resources

Simulation environment: a bookstore with bookshelves and tables as obstacles.

[mlherd/Dataset-of-Gazebo-Worlds-Models-and-Maps](https://github.com/mlherd/Dataset-of-Gazebo-Worlds-Models-and-Maps)

